

LD7 Full-Trajectory Sparse-Density Tractability Specification

Deterministic spread subsampling, bounded block streaming, and exact source-group batching

mdstats development specification

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Contents

LD7 — Full-Trajectory Sparse-Density Tractability	1
1. Objective	1
2. Preserved estimator	2
3. Public options	2
3.1 Spread-estimation options	2
3.2 Sparse execution options	2
4. Deterministic stratified-random spread sample	2
4.1 Scientific policy	3
4.2 Reliability gate	3
5. Bounded block-streaming convolution	3
5.1 Pre-LD7 failure	3
5.2 Two-pass block algorithm	3
5.3 Reduction policy	3
6. Exact source-group batching	3
7. Group-batched Phase-B planning	4
8. Data and serialization contracts	4
9. Resource and failure policy	4
10. Na-LTA 1,300-frame acceptance benchmark	5
11. Required tests	5
12. References	5

LD7 — Full-Trajectory Sparse-Density Tractability

Package target: mdstats 0.19.53a0

Status: normative and implemented

Scope: periodic atomic, framework-vertex, and framework-edge density preparation

1. Objective

LD7 makes the completed single-level block-sparse density architecture practical for long, finely resolved trajectories without changing the scientific estimator.

The triggering case is a 1,300-frame, 168-atom Na-LTA trajectory. SD-controlled refinement produces logical grids as large as approximately 1121^3 . Sparse retained fields fit comfortably in memory, but two pre-LD7 algorithms did not:

1. the exact periodic-spread diagnostic evaluated every frame with a multi-start torus mean;
2. optimized sparse convolution generated arrays proportional to every source-node/stencil pair.

LD7 separates **grid estimation** from **density estimation**:

- a bounded, deterministic randomized frame subsample estimates positional spread;
- every selected trajectory frame still contributes to the final density;
- convolution and planning stream bounded source-group batches into sparse blocks.

No grid interval, bandwidth, Gaussian tail tolerance, density mass, or HDR definition is relaxed to satisfy resources.

2. Preserved estimator

For weighted periodic samples $(\mathbf{f}_s, w_s)$, the production field remains

$$\rho = \frac{1}{\Delta V} G_h * \left(\sum_s w_s \delta_{\mathbf{f}_s} \right),$$

where C_h is periodic trilinear cloud-in-cell deposition and G_h is the canonical discrete periodized Gaussian stencil. The final normalization satisfies

$$\sum_i \rho_i \Delta V = \sum_s w_s.$$

LD7 changes only how the spread diagnostic and sparse convolution are executed.

3. Public options

3.1 Spread-estimation options

```
DensityResolutionOptions(  
    spread_sampling_strategy="stratified_random", # or "all"  
    spread_sample_size=128,  
    spread_sample_seed=0,  
)
```

These options are also exposed by AtomicDensityOptions and FrameworkDensityOptions.

Constraints:

- spread_sample_size >= 2;
- spread_sample_seed is an integer;
- spread_sampling_strategy is "all" or "stratified_random";
- when the source frame count does not exceed the requested sample size, all frames are used.

3.2 Sparse execution options

```
DensityOptimizationOptions(  
    sparse_evaluation_mode="optimized",  
    sparse_pair_chunk_size=262_144,  
    sparse_group_batch_size=8,  
    cache_stencil_supports=True,  
)
```

sparse_group_batch_size bounds the number of independent source groups—normally atoms, framework vertices, or framework edges—processed together.

4. Deterministic stratified-random spread sample

Let a trajectory contain T ordered frames with normalized weights a_t , and let $m < T$ be the requested spread sample size. Divide the ordered frames into m contiguous, equal-count temporal strata

$$I_j = [e_j, e_{j+1}), \quad e_j = \left\lfloor \frac{jT}{m} \right\rfloor.$$

Within each stratum, draw one frame with probability proportional to its local frame weight. Assign the selected frame the complete stratum weight

$$A_j = \sum_{t \in I_j} a_t.$$

The expensive periodic-mean diagnostic is then evaluated on the m selected frames using weights A_j .

This is an adaptation of standard stratified random sampling [1]. The temporal strata, deterministic NumPy generator, seed policy, and transfer of the full stratum weight to one selected frame are mdstats-specific.

4.1 Scientific policy

The sample is used **only** to resolve the positional-spread statistic and hence the grid/bandwidth. The density samples remain the complete requested frame collection.

For reproducibility, field metadata records:

- source and sampled frame counts;
- sample fraction;
- sampled source-frame indices;
- strategy and seed;
- resulting item SDs and selected quantile.

4.2 Reliability gate

For the Na-LTA acceptance trajectory, 128 stratified-random frames must reproduce the all-frame or converged-reference SD quantile closely enough that the resolved grid differs by no more than one practical refinement step. Focused synthetic tests require the sampled reference SD to agree within 2.5%.

5. Bounded block-streaming convolution

5.1 Pre-LD7 failure

The earlier optimized path generated

```
target_flat = empty(kernel_pair_count)
pair_values = empty(kernel_pair_count)
```

before reduction. Peak memory therefore remained $O(P)$, where

$$P = N_{\text{CIC}} N_{\text{stencil}}.$$

For the 1,300-frame oxygen field, $P \approx 5.6 \times 10^8$, making the temporary arrays larger than the runtime memory budget.

5.2 Two-pass block algorithm

For one source-group batch:

1. generate at most `sparse_pair_chunk_size` periodic target pairs;
2. map targets to logical block IDs;
3. mark the exact active target-block set;
4. allocate only those block-value arrays;
5. replay the same bounded pair stream;
6. reduce each chunk directly into block-local scalar slots;
7. extract positive logical nodes in deterministic global order;
8. normalize to the exact batch measure.

Peak package-owned memory is bounded by

$$O(BV + P_c + N_s + N_k),$$

where B is the active block count, V the block volume, P_c the pair-chunk bound, N_s the occupied CIC-node count, and N_k the stencil support size. It is no longer proportional to all P pairs.

5.3 Reduction policy

Small and reference-scale fields preserve stable `numpy.add.at` accumulation in the declared stencil-major/source-major order. Large batches may use a bounded block-local `numpy.bincount` reduction when the complete active block-slot vector is below the certified bound. The implementation choice is recorded in metadata.

6. Exact source-group batching

Density evaluation is linear. Partition source groups into deterministic ascending batches S_b :

$$\rho = \sum_b \rho_b.$$

Each batch:

- uses the same logical grid, cell, Gaussian support, and tail tolerance;
- carries its exact original sample weights;
- is normalized to its own measure;
- is evaluated with bounded block streaming.

The batch fields are merged by stable logical-node index reduction and corrected once to the exact total measure. Batch order is ascending source-group ID. The resulting metadata records batch count, source-group count, batch size, cumulative kernel pairs, peak batch workspace, and merge order.

Source groups are:

- selected atom index for atomic occupancy;
- framework vertex identity for vertex occupancy;
- framework edge or retained path identity for arc-length density.

7. Group-batched Phase-B planning

Transactional planning must predict the execution that realization will perform. LD7 therefore plans each deterministic group batch separately, unions exact target-node sets, and records:

- cumulative kernel-pair count;
- peak batch kernel-pair count;
- group count and batch count;
- final target-node and block-packing counts;
- bounded transient memory based on one batch, not all pairs.

The union is exact:

$$\mathcal{A} = \bigcup_b \mathcal{A}_b.$$

No scalar block values are allocated during planning.

8. Data and serialization contracts

PeriodicSpreadDiagnostics advances to

`mdstats.periodic-spread-diagnostic.v2`

and adds immutable sampling provenance. DensityOptimizationOptions adds `sparse_group_batch_size`, with canonical JSON round trips and a default of 8.

The framework-dynamics scene schema advances to

`mdstats.framework-dynamics-scene.v14`

because Phase-B plans and realized-field metadata now expose group-batched planning and spread-sampling provenance.

9. Resource and failure policy

The implementation must fail before unapproved allocation when any of these limits is exceeded:

- CIC contributions;
- stencil candidates;
- cumulative kernel pairs;
- pair-chunk workspace;
- active target blocks and block slots;
- planning bytes;
- retained field bytes.

Changing `sparse_group_batch_size` or `sparse_pair_chunk_size` may change performance and roundoff-scale reduction grouping, but not the logical grid, kernel, total measure, source samples, or declared density units.

10. Na-LTA 1,300-frame acceptance benchmark

Configuration:

- first 1,300 frames of TRAJECTORY (3);
- $dt = 1$ fs;
- framework-registered coordinates;
- all frames used for density;
- 128 stratified-random frames used for SD estimation;
- SD quantile 0.10;
- target artificial width/SD ratio 0.5;
- canonical discrete Gaussian, tail tolerance 10^{-3} ;
- 16^3 blocks, eight source groups per batch;
- 4 GiB workspace limit.

Measured density-preparation results on the validation runtime:

Species	Grid	Time	Cumulative kernel pairs	Integral
Na	525^3	13.74 s	96,251,957	24
Si	1121^3	33.42 s	164,431,073	24
Al	1038^3	16.56 s	158,246,794	24
O	690^3	60.15 s	558,840,991	96
Total	—	125.28 s	977,770,815	168

The benchmark includes density preparation, not three-shell isosurface extraction or browser serialization.

11. Required tests

- deterministic sampled indices for a fixed seed;
- complete temporal-stratum coverage;
- all-frame fallback when $T \leq m$;
- sampled SD agreement with the all-frame reference;
- exact CIC and active-node identities on reference fixtures;
- bounded scatter workspace independent of cumulative pair count;
- group-batched versus monolithic field agreement;
- group-batched planning union equal to monolithic target support;
- exact integrated measure after merging;
- deterministic source-group ordering;
- atomic, framework-vertex, and framework-edge execution;
- explicit `sampling_strategy="all"` compatibility;
- unchanged dense path and renderer contracts.

12. References

1. W. G. Cochran, *Sampling Techniques*, 3rd ed., Wiley, 1977. Stratified random sampling is adapted for weighted ordered trajectory frames.
2. R. W. Hockney and J. W. Eastwood, *Computer Simulation Using Particles*, Adam Hilger, 1988; reprint DOI: 10.1201/9780367806934. Periodic cloud-in-cell deposition.

The bounded block-streaming reduction, source-group batching, exact sparse merge, and transactional planning integration are original mdstats engineering designs.